Cloud Services Architecture

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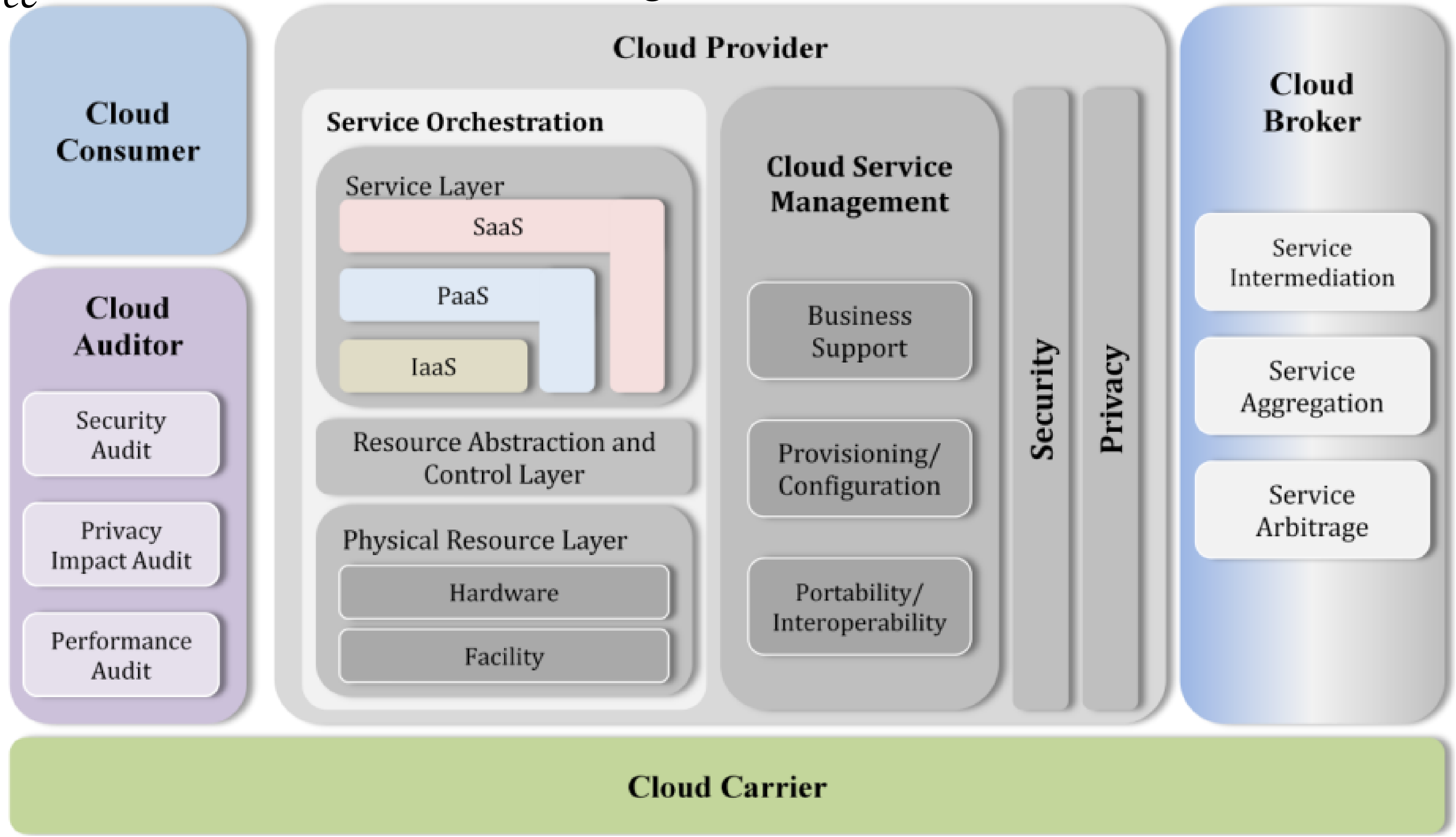
NIST Cloud Computing Architecture

Cloud Service Characteristics

- *On demand self-service*
- *Broad network access*
- *Resource pooling*
- *Rapid elasticity*
- *Measured service*

Understanding of the requirements, uses, characteristics and standards of cloud computing.

Cloud high-level architecture



NIST Cloud Computing Reference Architecture

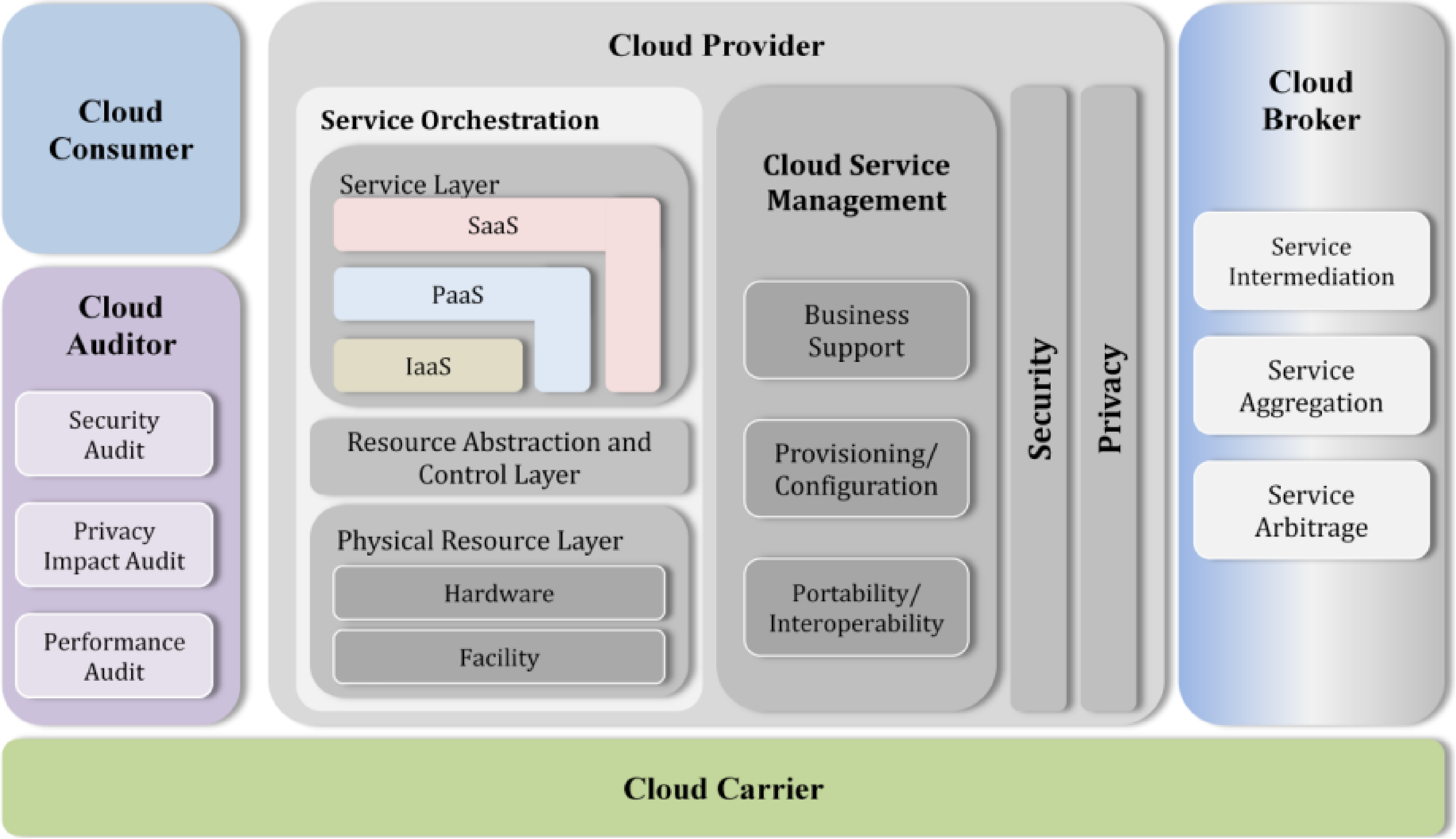
Cloud Conceptual Reference Model

Five Cloud Actors

1. *Cloud Consumer*
2. *Cloud Provider*
3. *Cloud Broker*
4. *Cloud Auditor*
5. *Cloud Carrier*

Actors with their roles, responsibilities, activities and functions in cloud computing.

Cloud high-level architecture



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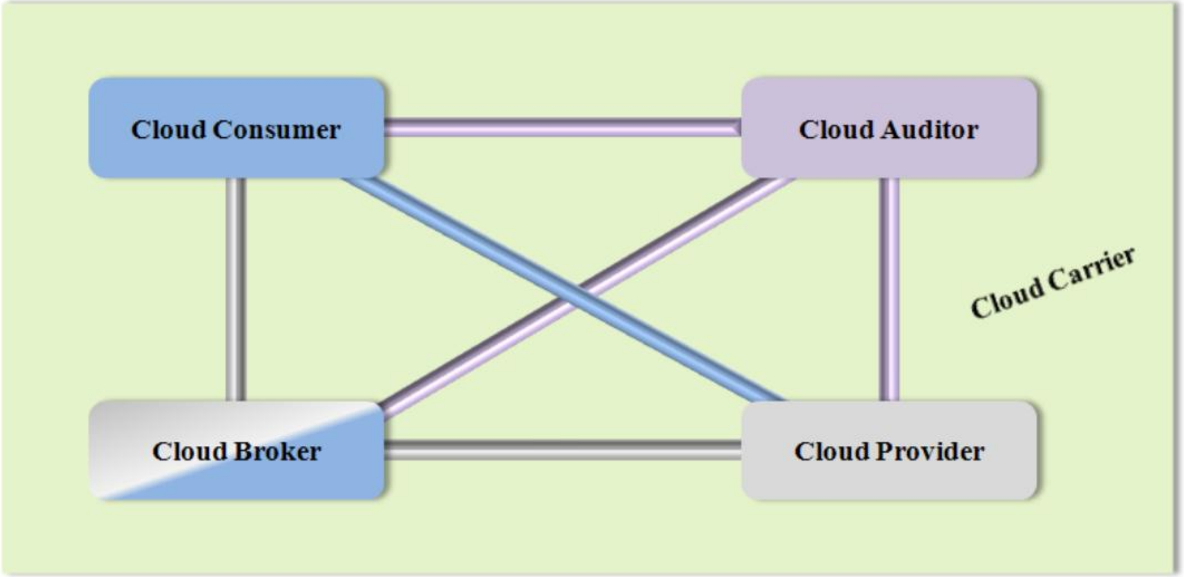
Actors in Cloud Computing

- **Cloud Consumer** A person or organization that maintains a business relationship with, and uses service from, *Cloud Providers*.
- **Cloud Provider** A person, organization, or entity responsible for making a service available to interested parties.
- **Cloud Auditor** A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
- **Cloud Broker** An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between *Cloud Providers* and *Cloud Consumers.*
- **Cloud Carrier** An intermediary that provides connectivity and transport of cloud services from *Cloud Providers* to *Cloud Consumers*.

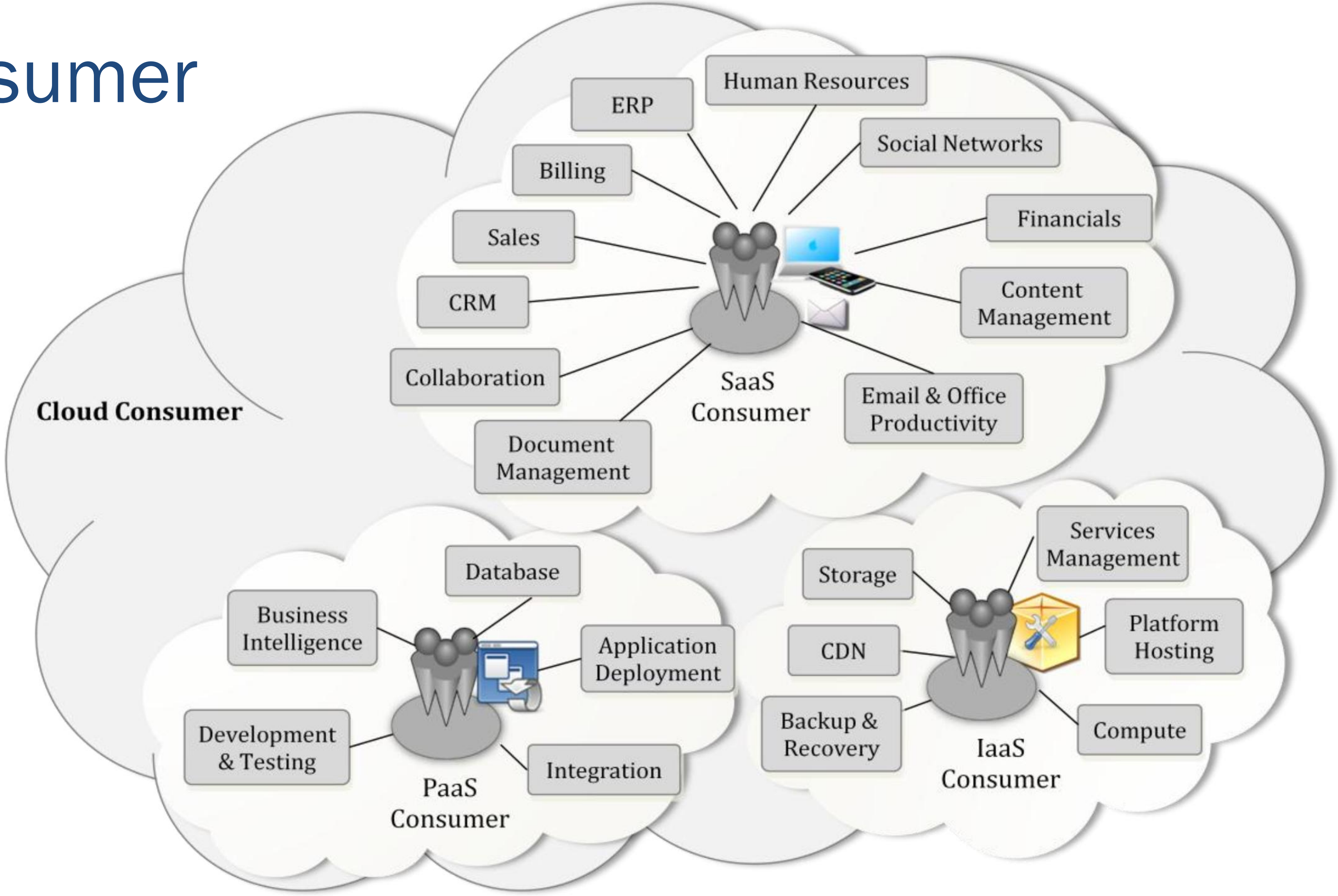


Cloud Consumer

- Cloud consumer browses & uses the service.
- Cloud consumer sets up contracts with the cloud provider.
- Cloud consumers need SLAs to specify the technical performance requirements fulfilled by a cloud provider.
- SLAs cover the quality of service, security, remedies for performance failures.
- A cloud provider list some SLAs that limit and obligate the cloud consumers by must acceptance.
- Cloud consumer can freely choose a cloud provider with better pricing with favorable conditions.
- Pricing policy and SLAs are non-negotiable.

Cloud Consumer

- SaaS Consumer
- PaaS Consumer
- IaaS Consumer



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Cloud Consumer

- SaaS consumers can be organizations that provide their members with access to software applications, end users who directly use software applications, or software application administrators who configure applications for end users.
- PaaS consumers can be application developers or administrators
  1. who design and implement application software
  2. application testers who run and test applications
  3. who publish applications into the cloud
  4. who configure and monitor application performance.
- IaaS consumer can be system developers, system administrators and IT managers who are interested in creating, installing, managing and monitoring services for IT infrastructure operations.

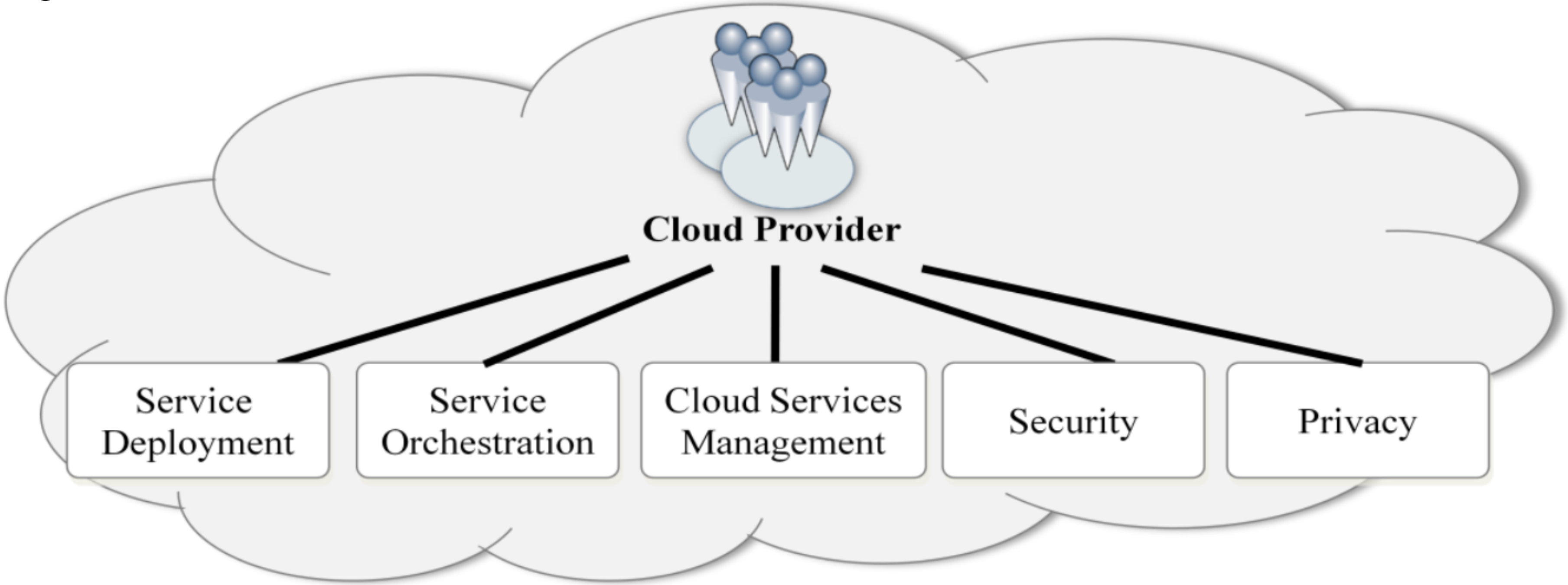
Cloud Consumer Billing

- SaaS consumers can be **billed** based on the number of end users, the time of use, the network bandwidth consumed, the amount of data stored or duration of stored data.
- PaaS consumers can be **billed** according to, processing, database storage and network resources consumed by the PaaS application, and the duration of the platform usage.
- IaaS consumer can be **billed** according to the amount or duration of the resources consumed, such as CPU hours used by virtual computers, volume and duration of data stored, network bandwidth consumed, number of IP addresses used for certain intervals.

Cloud Provider

Five major activities of Cloud Provider's

- *Service deployment*
- *Service orchestration*
- *Cloud service management*
- *Security*
- *Privacy*



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Cloud Provider

- Cloud Provider acquires and manages the computing infrastructure required for providing the services, runs the cloud software that provides the services, and makes arrangement to deliver the cloud services to the Cloud Consumers through network access.
- SaaS provider deploys, configures, maintains and updates the operation of the software applications on a cloud infrastructure. SaaS provider maintains the expected service levels to cloud consumers.
- PaaS Provider manages the computing infrastructure for the platform and components (runtime software execution stack, databases, and other middleware).
- IaaS Cloud Provider provides physical hardware and cloud software that makes the provisioning of these infrastructure services, for example, the physical servers, network equipments, storage devices, host OS and hypervisors for virtualization.

Cloud Broker

- Integration of cloud services can be complex for consumers. Hence cloud broker, is needed.
- Broker manages the use, performance and delivery of cloud services and negotiates relationships between cloud providers and cloud consumers.
- In general, a cloud broker can provide services in three categories:
  - *Service Intermediation*: Broker enhances a service by improving capability and providing value-added services to consumers. The improvement can be managing access to cloud services, identity management, performance reporting, enhanced security, etc.
  - *Service Aggregation*: Broker combines and integrates multiple services into one or more new services. The broker provides data integration and ensures the secure data movement.
  - *Service Arbitrage*: It is similar to service aggregation with the flexibility to choose services from multiple agencies. For example, broker can select service with the best response time.

Scenarios in Cloud: 1

1. Cloud consumer interacts with the cloud broker instead of contacting a cloud provider directly.
2. The cloud broker may create a new service (mash up) by combining multiple services or by enhancing an existing service.
3. Actual cloud providers are invisible to the cloud consumer.

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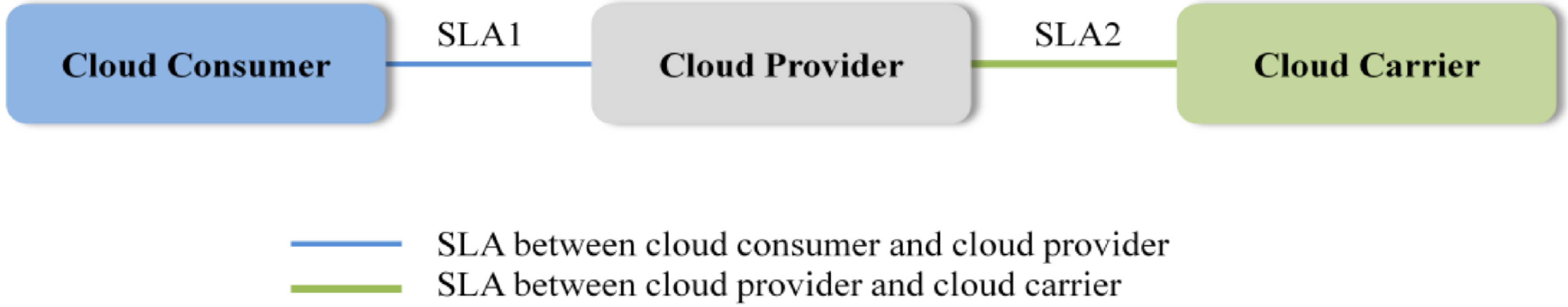
Cloud Carrier

- Cloud carriers provide access to consumers through network, telecommunication and other access devices.
- For example, cloud consumers can obtain cloud services through network access devices, such as computers, laptops, mobile phones, mobile internet devices (MIDs), etc.
- The distribution of cloud services is normally provided by network and telecommunication carriers or a *transport agent*, where a transport agent refers to a business organization that provides physical transport of storage media such as high-capacity hard drives.
- Cloud provider can set up SLAs with a cloud carrier to provide services consistent with the level of SLAs offered to cloud consumers.

Scenarios in Cloud: 2

1. Cloud carriers provide the connectivity and transport of cloud services from cloud providers to cloud consumers.
2. Cloud provider participates in and arranges for two unique service level agreements (SLAs), one with a cloud carrier (e.g. SLA2) and one with a cloud consumer (e.g. SLA1).
3. A cloud provider may request cloud carrier to provide dedicated and encrypted connections to ensure the cloud services (SLA's).



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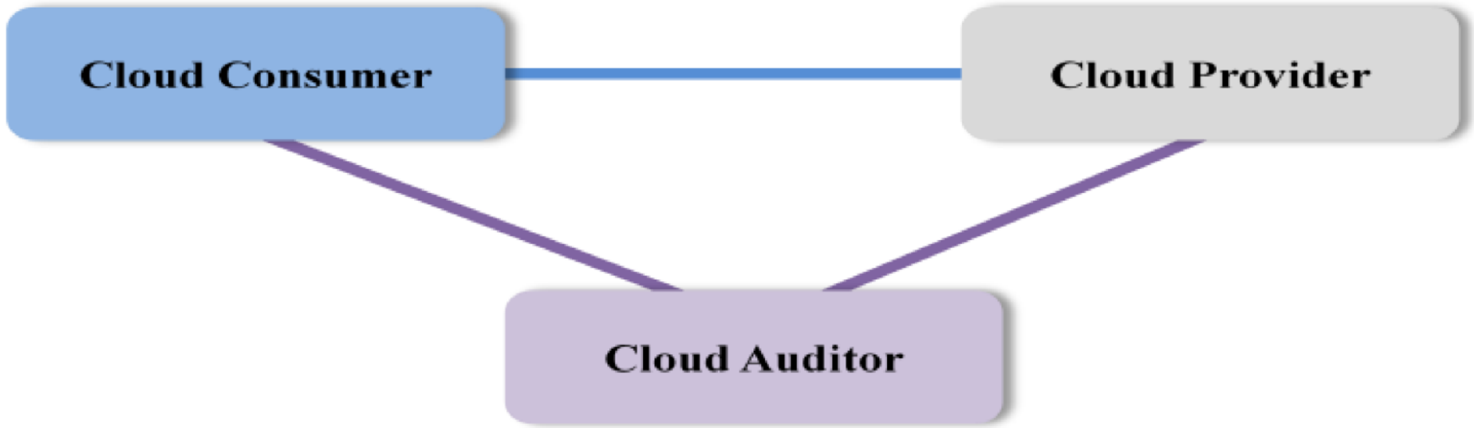
Cloud Auditor

- Audits are performed to verify conformance to standards.
- Auditor evaluates the security controls, privacy impact, performance, etc.
- Auditing is especially important for federal agencies.
- Security auditing, can make an assessment of the security controls to determine the extent to which the controls are implemented correctly, operating as intended, and producing the desired outcome. This is done by verification of the compliance with regulation and security policy.
- Privacy audit helps in Federal agencies comply with applicable privacy laws and regulations governing an individual's privacy, and to ensure confidentiality, integrity, and availability of an individual's personal information at every stage of development and operation.

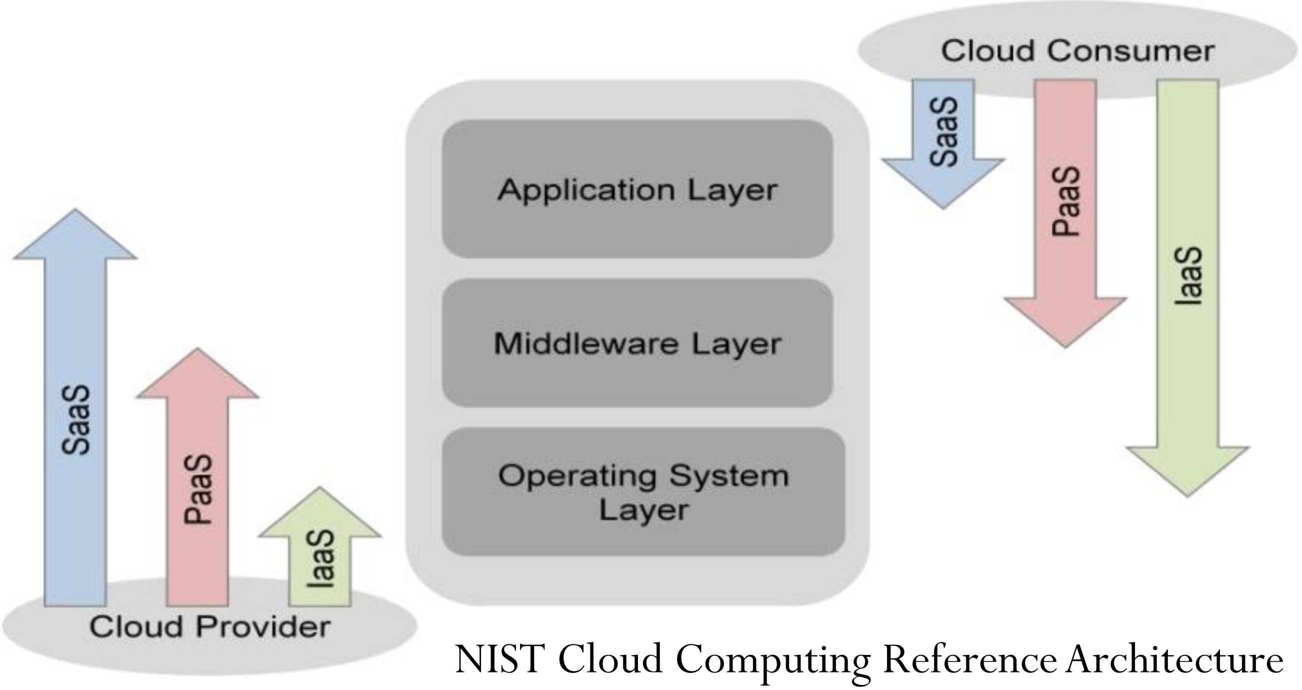
Scenarios in Cloud: 3

1. Cloud auditor conducts independent assessments for the operation and security of the cloud service.
2. The audit may involve interactions with both the Cloud Consumer and the Cloud Provider.



Scope of Control between Provider and Consumer

- Application layer are used by SaaS consumers, or installed/managed/ maintained by PaaS consumers, IaaS consumers, and SaaS providers.
- Middleware is used by PaaS consumers, installed/managed/maintained by IaaS consumers or PaaS providers. Middleware is hidden from SaaS consumers.
- IaaS layer is hidden from SaaS consumers and PaaS consumers. Consumers have freedom to choose OS to be hosted.



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Examples of Cloud Consumers

- **SaaS Consumers:** Students and teachers using **Google Docs, Sheets, and Classroom** for assignments, collaboration, and online teaching; users accessing **LinkedIn, Facebook, Zoom, Gmail** without managing software.

- **PaaS Consumers:** Developers and data scientists using **Google Colab, Docker, Kubernetes, and AWS Elastic Beanstalk** to develop, train, and deploy AI models and cloud applications without managing servers.

- **IaaS Consumers:** AI startups and enterprises renting **NVIDIA GPU servers (GPUaaS), AWS EC2, and Google Compute Engine** to train large language models and deploy scalable cloud services.

Examples of Cloud Providers

- **SaaS Providers: Google (Docs, Classroom), Microsoft (Office 365), Meta (Facebook, Instagram), Zoom** providing fully managed applications over the internet.
- **PaaS Providers: Google Colab, AWS Elastic Beanstalk, Azure App Services, Docker, Kubernetes platforms** providing development, deployment, and AI execution environments.
- **IaaS Providers: AWS, Google Cloud, Microsoft Azure, Oracle Cloud, IBM Cloud** providing virtual machines, GPU clusters, storage, and high-speed networking.
- **AI Infrastructure Providers: NVIDIA GPU Cloud, Google TPU Pods, AWS H100 clusters** providing large-scale AI compute and GPU-as-a-Service (GPUaaS).

Examples of Cloud Carriers, Brokers & Auditors

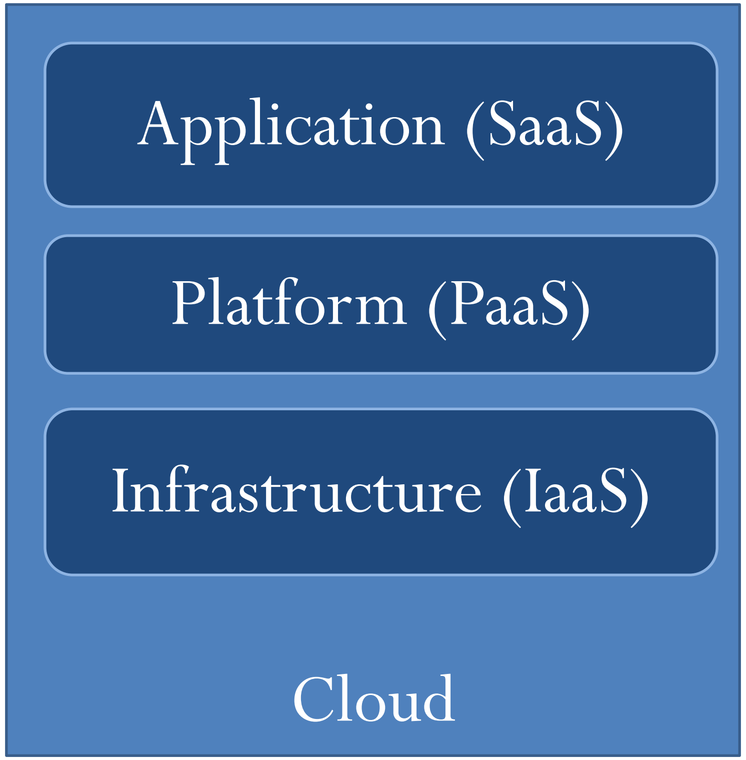
- **Cloud Carriers: Jio, Airtel, BSNL, AT&T, Verizon** providing fiber-optic, broadband, and 5G connectivity linking users to cloud data centers.

- **Cloud Brokers: Flexera (RightScale), CloudHealth, Spot.io** enabling multi-cloud management, workload optimization, and cost control.

- **Cloud Auditors: Deloitte, PwC, KPMG** auditing cloud platforms for security, compliance, data privacy, and regulatory standards (SOC2, ISO 27001, GDPR).
  - SOC (System and Organization Controls)
  - ISO (International Organization for Standardization)
  - GDPR (General Data Protection Regulation)

https://sites.google.com/site/animeshchaturvedi07/academic-teaching/cloudcomputing

Cloud Services

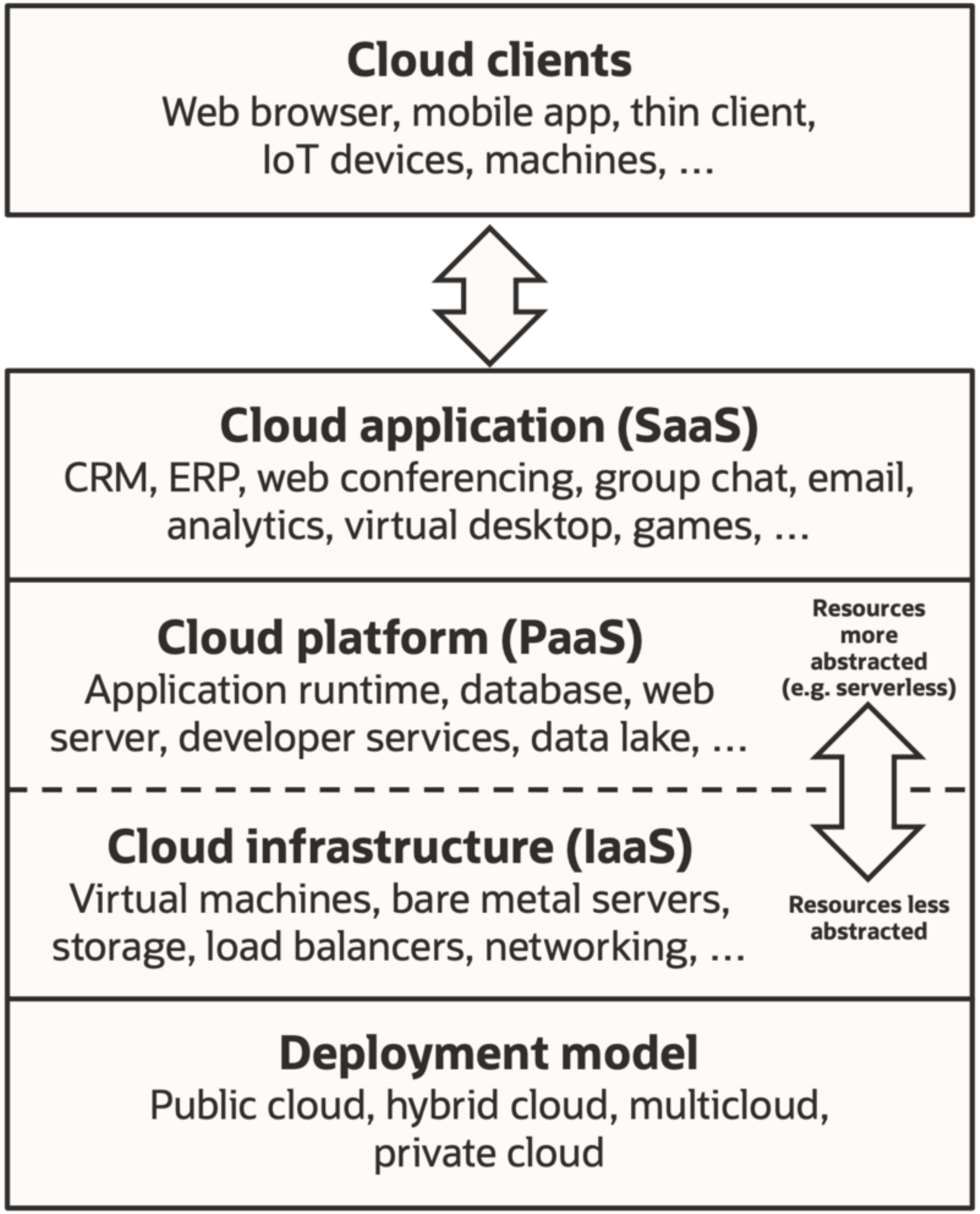


Classification and Vendors

- Infrastructure as a service (IaaS) offers hardware environment services. Kind of storage services (database or disk storage) or virtual servers. Some vendors are Amazon EC2, Amazon S3,Rackspace Cloud Servers and Flexiscale.
- Platform as a Service (PaaS) offers development (platform) environment services. PaaS offered by different vendors are generally not compatible. Typical players in PaaS are Google Application Engine, Microsofts Azure, Salesforce.com .
- Software as a service (SaaS) offers software. Application are based on 'pay-as-you-go' basis. Vendors Salesforce.coms in Customer Relationship Management (CRM), Gmail and Hotmail, Google docs and BPOS (Business Productivity Online Standard Suite) a MS Office online.

Different “X as a Service”

- Application-as-a-service
- Platform-as-a-service
- Integration-as-a-service
- Storage-as-a-service
- Database-as-a-service
- Information-as-a-service
- Process-as-a-service
- Security-as-a-service
- Testing-as-a-service
- Infrastructure-as-a-service
- Management/Governance-as-a-service



Example of Cloud Applications

- Some of the important applications
  - Twitter - twitter.com
  - Facebook - facebook.com
  - Google Apps for Business - google.com (text documents, spreadsheets, slide shows, Google Docs and many more)
  - Skype - skype.com
  - SalesForce https://www.salesforce.com/
- There are many cloud application please refer to http://en.wikipedia.org/wiki/Category:Cloud_applications

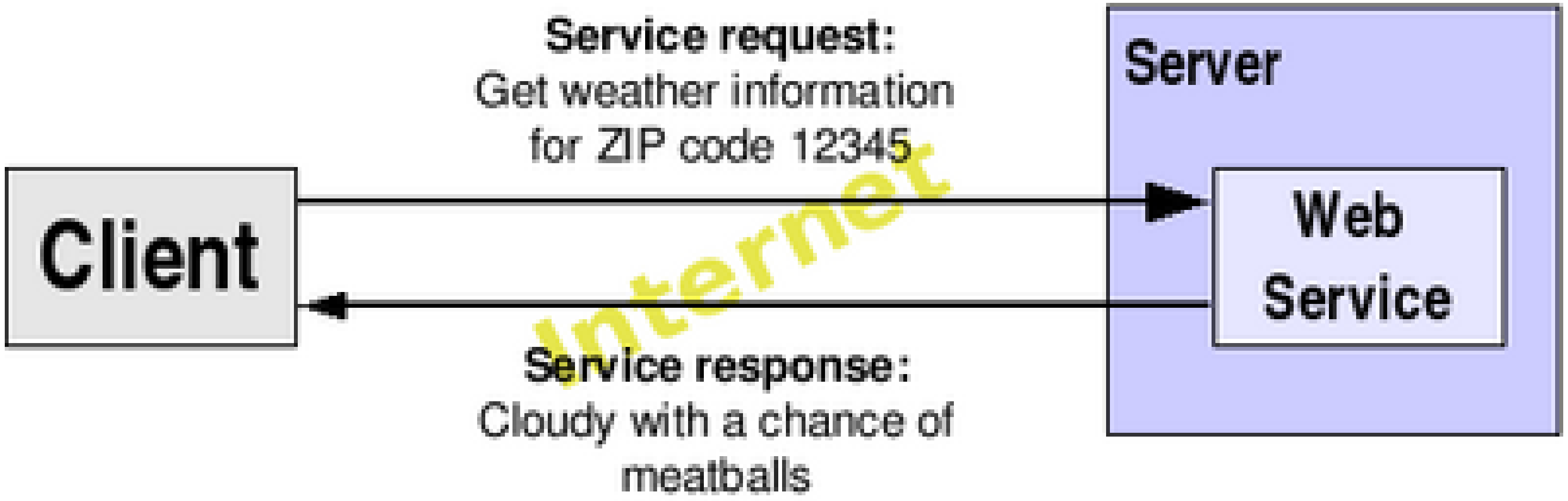
Web Services in Grid and Cloud Computing

How Services are being involved in
Grid and Cloud computing architecture.

https://sites.google.com/site/animeshchaturvedi07/academic-teaching/cloudcomputing

URL V/S URI

- Websites for humans, Web Services for software
- http://webservices.mysite.com/weather/us/WeatherService
- Web Services use HTTP as the application layer protocol to exchange SOAP messages over the network.

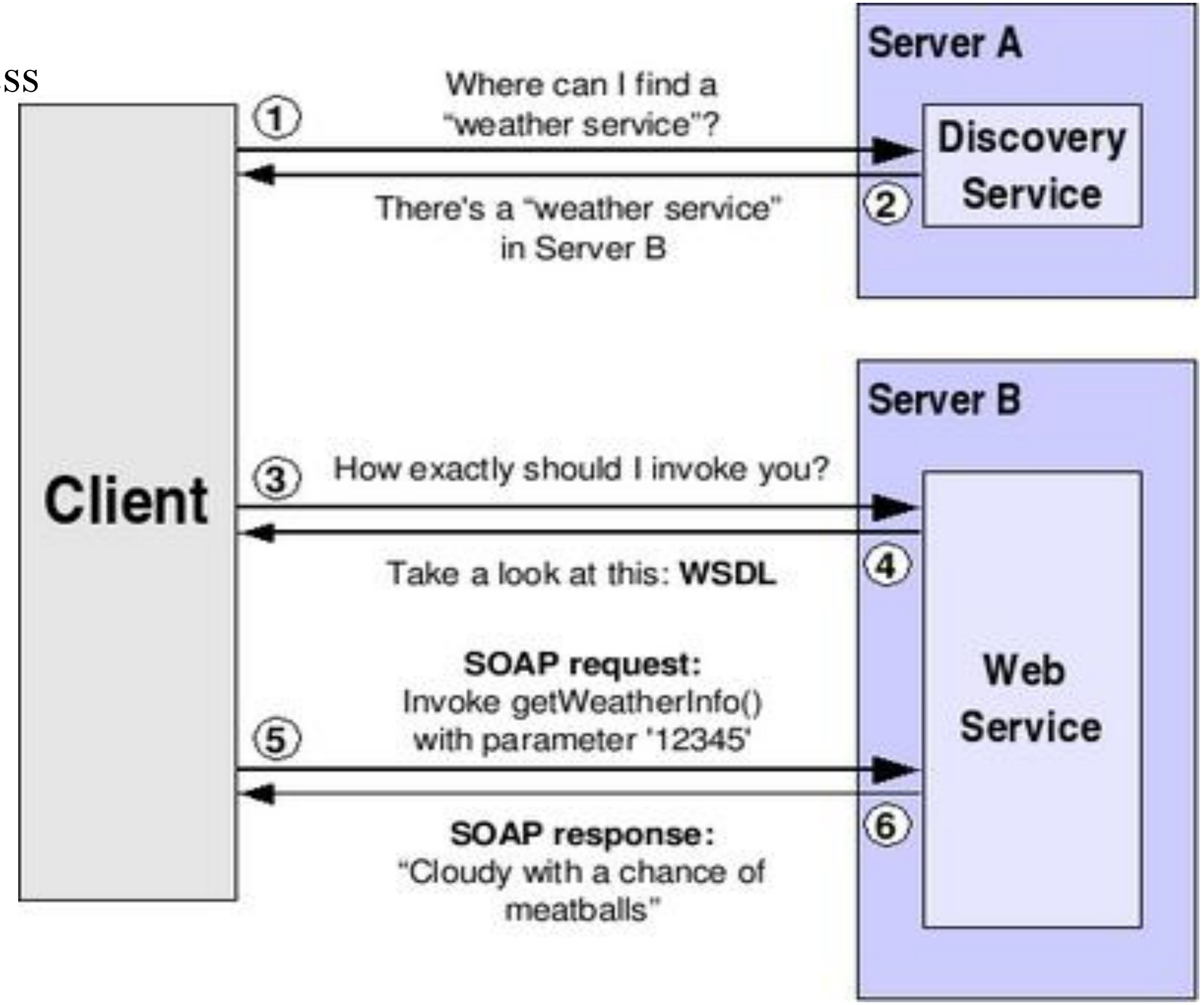


Borja Sotomayor, "The Globus Toolkit 4 Programmer's Tutorial,"

Web Service Invocation

Sequence of Web service invocation process

- Discovery service
- Discovery service will reply
- Web Service Descriptive Language (WSDL)
- SOAP request
- Method invocation
- SOAP response

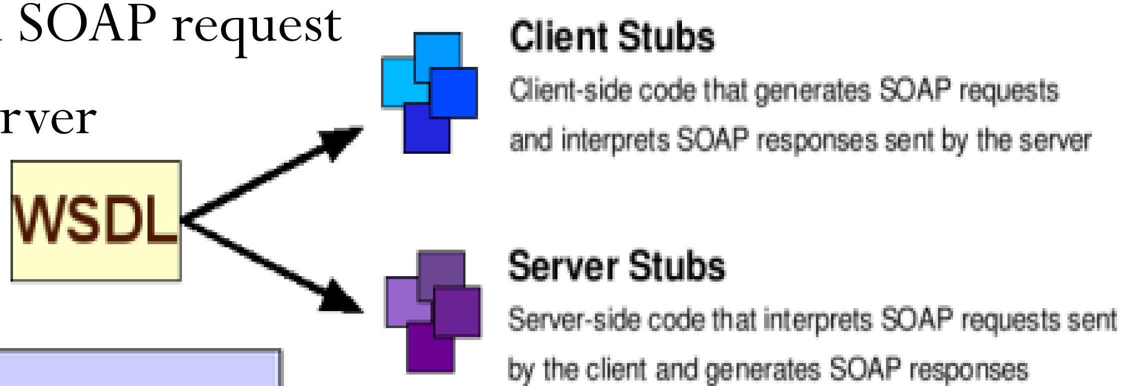


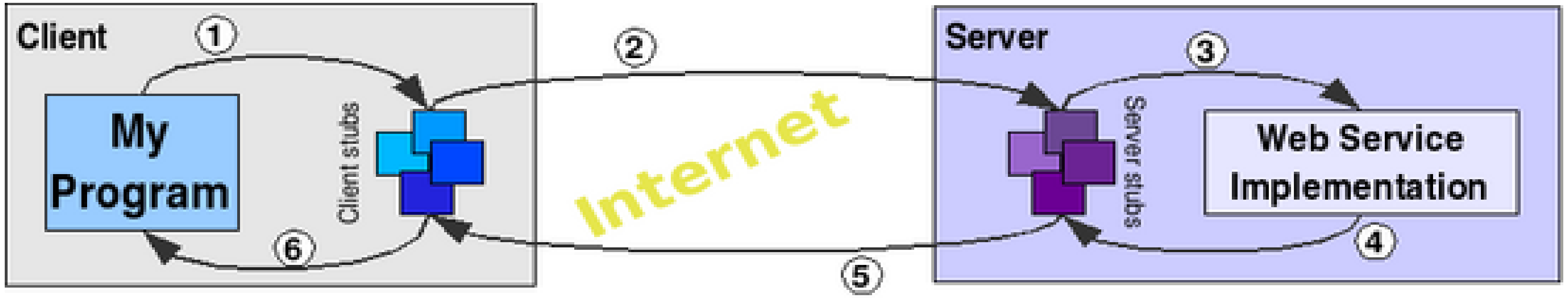
Borja Sotomayor, "The Globus Toolkit 4 Programmer's Tutorial,"

Web Service Invocation

- A stub is an API that handles Marshalling/Unmarshalling.
- Marshalling also refers to serializing and Unmarshalling also refers to deserializing.

1. SOAP request then Marshalling or serializing by Client
2. HTTP protocol
3. Unmarshaling or deserializing at Server then SOAP request
4. SOAP response then again Marshalling by Server
5. HTTP protocol
6. Unmarshaling then SOAP response at Client

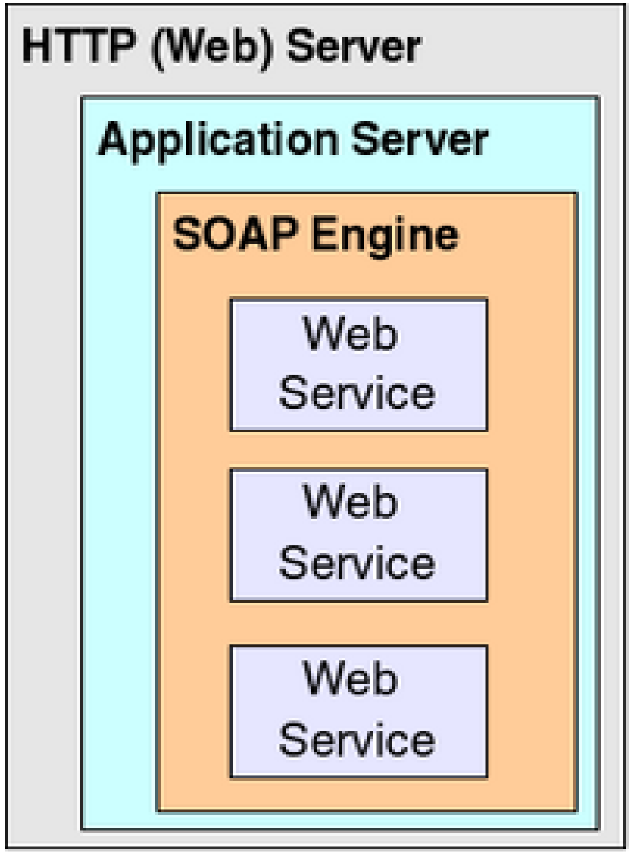




Borja Sotomayor, “The Globus Toolkit 4 Programmer's Tutorial,”

Web Service Invocation

- Web service
- SOAP engine
- Application server
- HTTP Server



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REST API

Representational State Transfer (REST)

- *Representational State Transfer* (REST) was introduced and defined in 2000 by Roy Fielding in his doctoral dissertation.
- The REST principles: "HTTP Object Model" begin and design *Uniform Resource Identifiers* (URI) standards in 1994.
- Intended to invoke an image: it is a network of Web resources (a Virtual State Machine) where the user progresses through the application by selecting resource identifiers and resource operations such as GET or POST (application state transitions),
- Resulting in the next resource's representation (the next application state) being transferred to the end user for their use.

https://en.wikipedia.org/wiki/Representational_state_transfer

Representational State Transfer (REST)

- WWW: "Web resources" are documents or files at URLs.
- Web 2.0: “Web Resources” are generic and abstract entities, or actions that can be identified, named, addressed, handled, or performed in many way on the internet as URI or URL.
- requests to a resource's URI:
  - a response with a payload formatted in many ways
  - e.g., HTML, XML, JSON, or some other format.
- The response given alternate resource state,
  - provide hypertext links to alternate related resources.

https://en.wikipedia.org/wiki/Representational_state_transfer

Representational State Transfer (REST)

HTTP Method	Operations for resource management
GET	*Retrieve* the URIs and representation of resource in the response body.
POST	*Create* a resource using the instructions in the request body. The URI of the created resource is *automatically assigned* and returned in the response *Location* header field.
PUT	*Replace* all the representations of the resources with the representation in the request body, or *create* the resource if it does not exist.
PATCH	*Update* all the representations of the resources of the resource using the instructions in the request body, or *may create* the resource if it does not exist.
DELETE	*Delete* the representations of the resources.

https://en.wikipedia.org/wiki/Representational_state_transfer

Web 2.0 and Mash-up

Web 2.0

Features

- Web 2 based Homepage
- Search
- Really Simple Syndication (RSS)
- Chats, Sharing
- Personalized web sites,
- Mashups
- POD Casting, Video Casting, You Tube
- Focus on community, online collaboration, tagging

Mashup

- A mashup, is a phenomenon to create new services on web, with the combination of presentation, data, or functionality from two or more resources.
- What are the benefits?
  - Innovation, new business insights, increase agility, speed up the development, reduce development costs, easy & fast integration, widgets are readily available, reuse software, immediate benefit, less cost, rich ecosystem, may not be original raw source data, produce enriched results of data & services
- Mashup is divided into three layers with the following technologies.
- **Presentation:** HTML, XHTML, CSS, Javascript, Ajax, API.
- **Web Services:** are XMLHTTPRequest, XML-RPC, JSON-RPC, SOAP, REST.
- **Data:** Sending, storing and receiving. The technologies used are XML, JSON, KML.

References

- NIST Cloud Computing Reference Architecture
- Borja Sotomayor, "The Globus Toolkit 4 Programmer's Tutorial,"
- https://en.wikipedia.org/wiki/Representational_state_transfer
- https://en.wikipedia.org/wiki/Web-oriented_architecture

תודה רבה
Hebrew

Ευχαριστώ
Greek

Спасибо
Russian

Danke
German

धन्यवाद:
Sanskrit

Merci
French

ধন্যবাদ
Bangla

நன்றி
Tamil

شكرا
Arabic

ಧನ್ಯವಾದಗಳು
Kannada

Thank You
English

നന്ദി
Malayalam

Grazie
Italian

多謝
Traditional Chinese

ధన్యవాదాలు
Telugu

આભાર
Gujarati

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Obrigado
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Japanese

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